

SECTION 23 10 00
FACILITY FUEL OIL SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

- A. Diesel fuel oil and unheated burner fuel oil tanks, piping, and accessories located outside, underground or aboveground as shown on contract drawings. Refer to contract drawings for type of fuel and for tank capacities.
- B. Tank fluid level monitoring and alarm systems.
- C. Leak detection system for tanks and underground piping.
- D. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

- A. Section 01 00 00, GENERAL REQUIREMENTS.
- B. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- C. Section 01 91 00, GENERAL COMMISSIONING REQUIREMENTS.
- D. Section 03 30 00, CAST-IN-PLACE CONCRETE.
- E. Section 05 50 00, METAL FABRICATIONS.
- F. Section 09 91 00, PAINTING.
- G. Section 13 05 41, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS.
- H. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.
- I. Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.
- J. Section 26 32 13, ENGINE GENERATORS.

1.3 APPLICABLE PUBLICATIONS

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.
- B. American Petroleum Institute (API):
 - 1. RP 1631-2001 Interior Lining and Periodic Inspection of Underground Storage Tanks
- C. American Society of Mechanical Engineers (ASME):
 - 1. B16.5-2017 Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard.
 - 2. B16.9-2018 Factory Made Wrought Buttwelding Fittings
 - 3. B16.11-2016 Forged Fittings, Socket-Welding and Threaded

4. B31.1-2018 Power Piping
- D. American Society for Testing and Materials (ASTM):
 1. A36/A36M-2017 Standard Specification for Carbon Structural Steel
 2. A53/A53M-2018 Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
 3. A105/A105M-2014 Standard Specification for Carbon Steel Forgings for Piping Applications
 4. A106/A106M-2015 Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
 5. A126-04(R2019) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
 6. A234/A234M-2015 Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
 7. B62-2017 Standard Specification for Composition Bronze or Ounce Metal Castings
 8. D2996-2015 Standard Specification for Filament-Wound "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe
- E. Federal Specifications (Fed. Spec.):
 1. A-A-60005-1998 Frames, Covers, Grating, Steps, Sump and Catch Basin, Manhole
- F. NACE International (NACE):
 1. SP0169-2013 Control of External Corrosion on Underground or Submerged Metallic Piping Systems
 2. 3/SSPC-SP6-2013 Commercial Blast Cleaning
 3. 4/SSPC-SP7-2006 Brush-off Blast Cleaning
- G. National Electrical Manufacturers Association (NEMA):
 1. 250-2014 Enclosures for Electrical Equipment (1000 Volts Maximum)
- H. National Fire Protection Association (NFPA):
 1. 30-2024 Flammable and Combustible Liquids Code
 2. 31-2024 Standard for the Installation of Oil-Burning Equipment
 3. 37-2024 Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines
 4. 70-2023 National Electrical Code (NEC)
- I. Steel Tank Institute (STI):
 1. F001-2017 Flameshield® Standard for Fire Tested Tanks
 2. F841-2010 Standard for Dual Wall Underground Steel Storage Tanks

3. F894-2018 ACT-100® Specification for External Corrosion Protection of FRP Composite Steel Underground Storage Tanks
 4. F941-2015 Fireguard: Specification for Fireguard Protected Aboveground Storage Tanks
 5. F961-2017 ACT-100U Specification for External Corrosion Protection of Composite Steel Underground Storage Tanks
 6. P3-2015 Specification and Manual for External Corrosion Protection of Underground Steel Storage Tanks
 7. R891-2006 Recommended Practice for Hold Down Strap Isolation
- J. Underwriters Laboratories Inc. (UL):
1. 58-2018 Standard for Steel Underground Tanks for Flammable and Combustible Liquids
 2. 142-2019 Standard for Steel Aboveground Tanks for Flammable and Combustible Liquids
 3. 971-2019 Standard for Nonmetallic Underground Piping for Flammable Liquids
 4. 1316-2018 Glass-Fiber-Reinforced Plastic Underground Storage Tanks for Petroleum Products, Alcohols, and Alcohol-Gasoline Mixtures
 5. 1746-2007(R2014) Standard for External Corrosion Protection Systems for Steel Underground Storage Tanks
 6. 2085-2019 Standard for Protected Aboveground Tanks for Flammable and Combustible Liquids

1.4 SUBMITTALS

- A. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.
- B. Information and material submitted under this section shall be marked "SUBMITTED UNDER SECTION 23 10 00, FACILITY FUEL OIL SYSTEMS", with applicable paragraph identification.
- C. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.
- D. Fuel Piping:
 1. ASTM and UL compliance.
 2. Grade, class or type, schedule number.
 3. Manufacturer.

- E. Pipe Fittings, Unions, Flanges:
 - 1. ASTM and UL compliance.
 - 2. ASTM standards number.
 - 3. Catalog cuts.
 - 4. Pressure and temperature rating.
- F. Foot Valves, Check Valves, Overfill Prevention Valves:
 - 1. Catalog cuts showing design and construction.
 - 2. Pressure and temperature ratings.
 - 3. Pressure loss and flow rate data.
 - 4. Materials of construction.
 - 5. Accessories.
- G. Secondary Containment System for Fuel Piping:
 - 1. Sizes, materials, construction of containment system including end seals, sumps, coatings and pipe supports.
 - 2. Layout of system.
 - 3. Installation instructions.
 - 4. Design of cathodic protection system (steel casing).
- H. Pre-engineered Double Containment Piping Systems:
 - 1. ASTM and UL compliance.
 - 2. Grade, class or type, schedule number.
 - 3. Manufacturer.
 - 4. Complete factory engineered system design.
 - a. Shop drawings.
 - b. Calculations.
- I. Leak Detection System:
 - 1. Drawings, description and performance data on sensors, control units.
 - 2. Description of operation.
 - 3. Layout of system.
 - 4. Installation and operating instructions.
 - 5. Data on interconnecting wiring systems to be furnished.
- J. Tank Fluid Level Monitoring Instrumentation System:
 - 1. Drawings showing instruments and in-tank sensing units, with dimensions.
 - 2. Design and construction of all elements of system.
 - 3. Installation instructions.

- K. Tank and Piping Accessories: Design, construction, and dimensions of vent caps, fill boxes, fill caps, spill containers and other accessories.
- L. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:
 - 1. Include complete list indicating all components of the systems.
 - 2. Include complete diagrams of the internal wiring for each item of equipment.
 - 3. Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.
- M. Completed System Readiness Checklist provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.
- N. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.

1.5 QUALITY ASSURANCE

- A. Approval by Contracting Officer is required of products or services of proposed manufacturers, suppliers and installers, and will be based on Contractor's certification that:
 - 1. Manufacturers regularly and currently manufacture tanks, tank and piping accessories, tank fluid level monitoring and leak detection systems, and fuel quality management systems.
 - 2. The design and size of each item of equipment provided for this project is of current production and has been in satisfactory operation on at least three installations for approximately three years. Current models of fluid level and leak detection systems with less than three years' service experience are acceptable if similar previous models from the same manufacturer have at least three years' service experience.
- B. Apply and install materials, equipment and specialties in accordance with manufacturer's written instructions. Conflicts between the manufacturer's instructions and the contract drawings and specifications shall be referred to the COR for resolution. Provide copies of installation instructions to the COR two weeks prior to commencing installation of any item.

- C. All equipment shall be free from defects that would adversely affect the performance, maintainability and appearance of individual components or overall assembly.
- D. Secondary Containment Systems for Piping, Tank Level Monitoring Systems, and Leak Detection Systems: Authorized manufacturer's representatives shall provide onsite training of installers and supervision of the installation and testing of the equipment and systems to assure conformance to written instructions of manufacturers.
- E. Piping and equipment installation contractor shall be certified as acceptable by local and state pollution control authorities.
- F. Entire installation shall conform to requirements of local and state pollution control authorities.
- G. Pipe Welding: Conform to requirements of ASME B31.1. Welders shall show evidence of qualification. Welders shall utilize a stamp to identify their work. Unqualified personnel will be rejected.
- H. Where specified codes or standards conflict, consult the COR.
- I. Label of Conformance (definition): Labels of accredited testing laboratories showing conformance to the standards specified.
- J. Equipment and materials installed shall be compatible in all respects with other items being furnished and with existing items so that the result will be a safe, complete and fully operational system which conforms to contract requirements and in which no item is subject to conditions beyond its design capabilities.

1.6 AS-BUILT DOCUMENTATION

- A. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.
- B. Certification documentation shall be provided to COR 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip

points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

1.7 PERMITS

- A. Contractor shall obtain and complete all tank permit and registration forms required by governmental authorities.

PART 2 PRODUCTS

2.1 PIPING, VALVES, FITTINGS

- A. Fuel supply and return, tank fill, vents, sounding, and pump out.
- B. Steel Pipe and Fittings:
 - 1. Piping: Steel, seamless or electric resistance welded (ERW), ASTM A53/A53M Grade B or ASTM A106/A106M Grade B, Schedule 40.
 - a. Aboveground piping shall be painted. Refer to Section 09 91 00, PAINTING.
 - 2. Joints: Socket or butt-welded. Threaded joints are prohibited except at valves, unions and tank connections.
 - 3. Fittings:
 - a. Butt-welded joints: Steel, ASTM A234/A234M, Grade B, ASME B16.9, same schedule as adjoining pipe.
 - b. Socket-welded joints: Forged steel, ASME B16.11, 13,790 kPa (2000 psig) class.
 - 4. Unions: Malleable iron, 2070 kPa (300 psig) class.
 - 5. Companion flanges: Flanges and bolting, ASME B16.5.
 - 6. Welding flanges: Weld neck, ASME B16.5, forged steel ASTM A105/A105M, 1034 kPa (150 psig).
- C. Check Valves - Fuel Pump Suction.
 - 1. Pipe Sizes 50 mm (2 inches) and under: Rated for 1375 kPa (200 psig) water-oil-gas, swing-type, threaded ends, ASTM B62 bronze body. Provide union adjacent to valve.
- D. Foot Valves - Fuel Pump Suction: Double poppet, lapped-in metal-to-metal seats, double-guided stems, 20 mesh inlet screen, same size as fuel suction piping. Foot valve shall be removable to above grade through the tank manhole enclosure or through extractor fitting.
- E. Extractor Fittings: Arranged to permit removal of foot valves, overfill prevention valves, and other devices that are located below grade. Access point shall be through a cast iron fill box-type manhole located at grade. Provide extractor wrench.

- F. Anti-Siphon Valves: Brass body, two-way, one directional flow, hung piston type, normally-closed electrical solenoid valve. Seals shall be Viton material. NFPA 30 & 30A, UFC, BOCA, SBCCI/SFC, and PEI RP200 compliant, CSA Certified. No differential pressure shall be required to open the valve. Solenoid shall be housed in an integral, watertight, explosion-proof shell. Solenoid coil shall be continuous duty type.
- G. Solenoid Valves: Brass body, two-way, one directional flow, hung piston type, normally-closed electrical solenoid valve. Seals shall be Viton or teflon material. NFPA 30 & 30A, UFC, BOCA, SBCCI/SFC, and PEI RP200 compliant, CSA Certified. No differential pressure shall be required to open the valve. Solenoid shall be housed in an integral, watertight, explosion-proof shell. Solenoid coil shall be continuous duty type. Integral thermal relief to allow the valve to relieve expansion pressure in the opposite direction of flow.

2.2 PRE-ENGINEERED DOUBLE CONTAINMENT PIPING SYSTEMS (ABOVE GROUND)

- A. Double-containment piping system shall be factory-engineered and prefabricated for field installation.
- B. Carrier Piping:
 - 1. ASTM A53B Black Steel pipe, Seamless or ERW, in pre-cut lengths. Pipe 10" and smaller shall be Schedule 40.
- C. Inner Pipe Supports:
 - 1. All pipe shall be aligned and supported within the casing with centering supports spaced on approximately 10'0". The inner pipe shall bear directly on the support. The support shall be designed as to permit drainage and free air passage. Concrete type pipe supports will not be allowed. Supports will be manufactured in such a way that 1" EMT tube will be able to go through for leak detection pull rope capabilities.
 - 2. Internal EMT tube guides shall be provided for all fitting sections.
- D. Outer Containment Casing:
 - 1. Outer casing shall be black steel. Casing up through 24" shall be 10 gauge.
 - 2. The interior surface shall be smooth to permit free moisture drainage and removability of the inner assembly. The outer casing shall be sized to provide an adequate annular space between the

outer surface of the insulation material and the interior surface of the casing.

3. Exterior Coating:

a. The exterior surface will be coated with a two coat Fusion Bonded Epoxy system. The first coat will be green finish coat with a melting point of 500°F. No glasswrap or filler materials shall be used in the epoxy. All exterior conduit surfaces shall be shotblasted prior to the coating being applied. The Fusion Bonded Epoxy shall conform to these ASTM Standards:

- 1) ASTM D1763 Disclosure of properties of the epoxy Sub-components
- 2) ASTM G17 Penetration test
- 3) ASTM D1044 Abrasion resistance
- 4) ASTM D2370 Tensile strength
- 5) ASTM G14 Impact tests
- 6) ASTM G8 Salt crock
- 7) ASTM D968 Abrasion tests
- 8) ASTM D1002 Shear strength and adhesion
- 9) ASTM D659 Compressive strength
- 10) ASTM D257 Volume resistivity
- 11) ASTM D1000 Electric strength
- 12) ASTM G53 Weathering
- 13) ASTM B117 Salt fog

b. No asphalt, coal tar coating, FRP casing or any other type will be allowed.

c. The second layer will be compatible Fusion Bonded Epoxy coating that will provide mechanical protection to the first layer. The second layer of Fusion Bonded Epoxy will be applied no later than 5 seconds after the first layer has been applied so that it securely bonds to the first layer as both layers cure. The second layer must have an impact resistance of at least 160 lbs. per square inch as per ASTM G14-72. The Fusion Bonded Epoxy coating will be applied in a total thickness of no less than 20 mils. 2 layers required.

4. Outer conduit casing closures shall consist of 10 gauge steel suitably rustproofed and in cylindrical form with a single horizontal split and shall be field welded over adjacent units.

E. Weld Fittings:

1. All changes in direction shall be made with bent or weld fittings. Where tee branches are smaller than the mains they join, weld-onlets may be used. All weld fittings shall be long radius and shall be the same wall thickness as adjacent piping.

F. Leak Detection:

1. Leak detection sensors shall be provided as part of the pre-engineered system or separately from the double-containment piping system.
 - a. Leak detection sensors shall be compatible with the leak detection monitoring system specified elsewhere in this section.
2. Pull ports, guide tubing, and pull wires shall be provided in the containment system to facilitate installation of cable-type leak detection.

G. End Seals:

1. Terminal ends at ends of double containment sections and at locations shown on drawings shall be equipped with end seals consisting of a steel bulkhead plate welded to the pipe conduit.
2. End seals shall be made of 1/2" steel plate with drain or vent openings located diametrically opposite on the vertical center line of the mounting plate and shall be shipped to the job site with plugs in place. Terminate all conduits 2 inches beyond the inside face of building walls to protect any exposed piping insulation from damp wall condensation.

2.3 LEAK DETECTION SYSTEMS

- A. Automatic digital continuous monitoring systems responsive to the presence of water and hydrocarbons in the interstitial space of the double-wall tanks, in the tank manhole access enclosures, and in the secondary containment of fuel piping systems. System shall distinguish between hydrocarbon and water and identify location of leak as to individual tank and piping system. System may be combined with tank fluid level monitor and alarm system specified in paragraph, TANK FLUID LEVEL MONITOR AND ALARM SYSTEM.
- B. Functions and Arrangement:
 1. Single control station to monitor all sensing probes.
 2. Visual indicator to monitor and identify leaks as water or hydrocarbon and location.

3. Indicators showing system status including faults and alarms.
4. On board printer that provides complete reports of all system functions upon command.
5. Panel circuit test button.
6. 95 dB audible alarm with silencing control to sound when leak is detected.
7. Eight-hour memory backup system with battery.
8. NEMA 250 Type 4 cabinet.
9. UL or other accredited testing laboratory listing.
10. Modbus or BACNet communications to DDC system to indicate system in service and alarm conditions.

C. Sensors:

1. Designed for required locations including: Insertion between walls of double-wall tanks, in sumps in double-wall piping systems and in tank manhole enclosures. Sensing points shall be at lowest point of each tank or sump. Intrinsically safe design.
2. Sensing units shall detect presence of water and a minimum 3.2 mm (1/8 inch) thick layer of hydrocarbon on surface of water and minimum 50 mm (2 inch) thickness of hydrocarbon in area that has no water present.
3. Sensors shall be arranged to allow replacement of individual sensors without disturbing other portions of leak detection system or fuel storage and piping system. Underground sensors shall be accessed through caps as grade.
4. Materials of construction shall be non-corroding.
5. Transmit status signal to control unit.
6. Sensors for double-containment piping systems shall be cabletype.

D. Components:

1. Provide manholes at grade for each sensor cap similar in construction to fill boxes. Manholes shall be cast iron, quick-opening cover, watertight, minimum size necessary to accommodate sensor caps. Provide identification plates, similar to those specified for fill points, labeled "MONITORING/OBSERVATION WELL-DO NOT FILL". Provide special tools if necessary for opening covers.
2. Sensor housings from tank and piping to grade shall be Schedule 40 PVC, or stainless steel.

3. Underground wiring between probes and control unit: Place in water-tight corrosion-resistant conduit system conforming to Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

2.4 TANK FLUID LEVEL MONITOR AND ALARM SYSTEMS

- A. Digital systems for central monitoring of fuel and water levels in all fuel oil storage tanks in the project. High and low level visual and audible alarms. Volumetric tank-tightness testing. Complete with all transducing, transmitting, and receiving devices. On board printer to provide complete report of all system functions upon command. System may be combined with leak detection system specified in paragraph, LEAK DETECTION SYSTEMS.
- B. Fluid Level Monitor:
 1. Digital continuous readout, showing tank oil and water levels in gallons, smallest reading one gallon. Provide identification of product measured, measuring units, and the tank number.
 2. Tank and fuel characteristics contained in preprogrammed non-volatile field-replaceable databases. Protected power supply.
- C. High and Low Fluid Level Alarm System:
 1. Automatic continuous on-line monitoring of all tanks.
 2. Visual and audible indicators combined with fluid level monitor. Identify the tank that is in alarm condition.
 3. Manual alarm test and silencing controls.
 4. Low level alarm actuation adjustable 0-25 percent of tank capacity. High level alarm actuation adjustable 75-100 percent of tank capacity.
- D. Locate all indicators, selector switches, alarms on face of wall-mounted NEMA 250, Type 4 panel.
- E. Remote Alarm Annunciator:
 1. Visual and audible high-level alarms adjacent to tank fill box locations. Locate in NEMA 250 Type 4X weatherproof exterior wall or pole-mounted panels.
 2. Alarm shall include flashing red light with 180-degree visibility for each tank and 95 dB horn or 100 mm (4 inch) diameter bell. Provide alarm silence control.
 3. Provide identification sign: "WHEN ALARM SOUNDS - FUEL TANK FILLED TO CAPACITY - DO NOT OVERFILL".

- F. Modbus or BACNet communication to DDC system to indicate tank fluid level and alarm conditions. Telephone modem communication capability.
- G. System Performance: Accuracy plus or minus 2.5 mm (0.10 inch) of fluid height in inventory mode and 0.25 mm (0.01 inch) in leak detection mode. Automatic compensation for fluid temperature changes. Volumetric tank tightness sensitivity of 0.4 lph (0.1 gph).
- H. Sensors:
 - 1. Provide sensor types such as magnetostrictive, capacitance, float, hydrostatic and other types as necessary for the applications.
 - 2. Apply in accordance with manufacturer's instructions with provisions for easy future replacement without need for excavation.
 - 3. Provide for each hydrostatic sensor a constant flow differential pressure regulator and transmitter protected from fuel contamination. Air supply shall include filter and over-pressure protection. Provide desiccant-type dryer on air supply designed for removal of water vapor. Dryer rating, minimum 4.6 L/s (10 SCFM). Provide moisture indicator. Dryer may be deleted if air supply source has a refrigerated dryer.
 - 4. Float-type units shall be designed for installation and removal through a 100 mm (4 inch) diameter vertical pipe mounted in the top of the tank.
- I. Underground Wiring and Piping: Enclose in water-tight corrosion-resistant conduit system sized and arranged as recommended by system manufacturer and conforming to Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.
- J. Code Conformance: NFPA 70.

2.5 PUMPS

- A. Cast iron construction, positive displacement, internal or external gear type, close coupled configuration, TEFC motor, with integral relief valve. Provide with mechanical seals.
- B. Pumps shall be designed for fuel oil applications.
- C. Provide each pump with:
 - 1. Inlet strainer.
 - 2. Outlet check valve.
 - 3. Leak sensor.
 - 4. Inlet suction gauge.

5. Outlet pressure gauge.
 6. Motor disconnect switch.
 7. Mounting frame.
 8. Indoor rated enclosure - provide for all pumps installed inside building but not located inside enclosure of other equipment. For duplex pump sets a single enclosure may contain both pump assemblies.
 9. Outdoor rated enclosure - provide for all pumps installed outside of building but not located inside enclosure of other equipment. For duplex pump sets a single enclosure may contain both pump assemblies.
- D. Provide simplex or duplex pump assemblies where noted on drawings.

PART 3 EXECUTION

3.1 GENERAL

- A. If an installation is unsatisfactory to the COR, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 INSTALLATION AND TESTING, UNDERGROUND PIPING SYSTEMS

- A. Existing underground piping systems shall be pressure tested prior to connecting any new piping.
1. The inner pipes of this system shall be tested to 1-1/2 times the working pressure of the line. The outer casing shall be tested to locate leaks.
 - a. Report any leakage detected to COR.
- B. Pressure testing:
1. Pressure test existing carrier piping as noted above prior to making connections to new above-grade piping systems. Report any leakage detected to COR.
 - a. Working pressure: 50 psig
 - b. Test pressure: 75 psig
 - c. Test type: hydrostatic
 - d. Test duration: 1 hour
 2. Pressure test existing containment piping as noted above prior to making connections to new above-grade piping systems. Report any leakage detected to COR.
 - a. Test pressure: 5 psig

- b. Test type: air
- c. Test duration: 1 hour

3.3 INSTALLATION AND TESTING, ABOVE-GRADE DOUBLE-CONTAINMENT PIPING SYSTEMS

- A. The inner pipes of this system shall be tested to pressures specified below. The outer casing shall be tested to locate leaks.
 - 1. If leaks occur, they shall be repaired and the test repeated.
After approved by test, all field joints shall be coated by the contractor.
- B. Pressure Testing:
 - 1. Pressure test new carrier piping as noted above prior to making connections to existing underground piping systems. Repair all leaks and retest.
 - a. Working pressure: 50 psig
 - b. Test pressure: 100 psig
 - c. Test type: hydrostatic
 - d. Test duration: 1 hour
 - 2. Pressure test new containment piping as noted above prior to making connections to existing underground piping systems. Repair all leaks and retest.
 - a. Test pressure: 15 psig
 - b. Test type: air
 - c. Test duration: 1 hour
 - 3. Pressure test carrier piping after connections between new and existing piping systems are made. Repair all leaks and retest.
 - a. Working pressure: 50 psig
 - b. Test pressure: 75 psig
 - c. Test type: hydrostatic
 - d. Test duration: 1 hour
 - 4. Pressure test containment piping after connections between new and existing piping systems are made. Repair all leaks and retest.
 - a. Test pressure: 5 psig
 - b. Test type: air
 - c. Test duration: 1 hour
- C. After tests all exposed closures shall be covered in the field with a two part epoxy and left to dry and then polyethylene heat shrink material with minimum thickness of 60 mils.

3.4 INSTALLATION AND TESTING, LEAK DETECTOR SYSTEMS FOR TANKS AND PIPING

- A. Wiring shall conform to NFPA 70.
- B. Locate control monitor panels 1500 mm (5 feet) above the floor on inside wall of boiler room, generator room or garage, depending on type of fuel tank served, unless shown otherwise.
- C. Test operation of each probe, and monitoring system with fuel and water. If type of probe utilized is damaged by exposure to fuel, provide temporary probe for testing monitoring system.

3.5 INSTALLATION, TANK FLUID LEVEL INDICATOR AND ALARM SYSTEM

- A. Wiring shall conform to NFPA 70.
- B. Locate level indicator and alarm panel 1500 mm (5 feet) above the floor on inside wall of boiler room, generator room or garage, depending on type of fuel tank served, unless shown otherwise.
- C. Locate remote high-level alarm on exterior wall or pole in view of tank fill point, 2400 mm (8 feet) above grade.

3.6 VALVES

- A. Valve solenoids shall be mounted with solenoid vertical and upright.

3.7 STARTUP AND TESTING

- A. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.
- B. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.
- C. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with COR and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

3.8 COMMISSIONING

- A. Provide commissioning documentation in accordance with the requirements of Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.
- B. Components provided under this section of the specification will be tested as part of a larger system.

3.9 DEMONSTRATION AND TRAINING

- A. Provide services of manufacturer's technical representative for 8 hours to instruct each VA personnel responsible in operation and maintenance of the system.
- B. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.

- - - E N D - - -

- - - This Page Left Intentionally Blank - - -